RAJAT MURHE

Final Year Computer Science Undergraduate
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PROFILE

Final year Computer Science undergraduate (BTech, AI and Machine Learning, VIT Chennai) with three internships, who builds **RAG and document intelligence** systems: contract review where every answer shows its exact source quote, an enterprise assistant that only answers from documents a person may see, and semantic search on a live product. I test for missing evidence, conflicting clauses and prompt injection.

EXPERIENCE

AI Product Growth and Strategy Intern, DCS AI Technologies Sep 2026 to Present

- Integrated **third party APIs** into AI powered product workflows, connecting external services with application features and validating end to end functionality.
- Worked on a platform's multimodal AI capabilities, testing and validating image, video, audio and product ad generation workflows, including API responses and **credit consumption**.
- Contributed to an AI connector platform's integration workflows by testing connectors, dashboard operations, execution flows and **role based permissions**.
- **Debugged** integration issues and documented test cases and findings to improve product reliability and release readiness.

AI/ML Intern, Linkific | Remote Jun 2026 to Aug 2026

- Built the **backend** for a Secure Enterprise AI Assistant that answers questions from company documents, using FastAPI, RAG on PostgreSQL with pgvector and a Groq LLM.
- Added **role based access control** to the API so people only get answers from documents they are allowed to see, and helped with troubleshooting and debugging for users.

AI Intern, Rungta Ventures AI Jan 2026 to May 2026

- Built a **semantic search** feature on MyCA using embeddings, and wrote Python scripts with NLTK and spaCy that validate uploaded documents, with a team of around ten people on a live product that uses Firebase for OTP login.

PROJECTS

ContractIQ, AI Contract Review Assistant | GitHub | Live
Next.js* | *React* | *TypeScript* | *Python* | *FastAPI* | *SQLite* | *OpenAI compatible APIs* | *Render

- Built and deployed a web app where users upload PDF, DOCX or TXT contracts, ask questions in plain language and see the **exact source quote** behind each answer, checked against the contract text.
- Wrote automated tests and evaluations for missing evidence, conflicting clauses and prompt injection, with 23 passing frontend tests.

AI Job Assistant, Resume and Job Match Tool | GitHub Jan 2026 to Aug 2026
Next.js* | *React* | *TypeScript* | *Redux Toolkit* | *FastAPI* | *PostgreSQL* | *pgvector* | *n8n* | *Gemini* | *Ollama* | *OpenAI APIs

- Built an AI job search assistant that reads a resume and a job description, shows how well they match with **evidence** for each matched and missing skill, and drafts tailored resumes, cover letters and emails.
- Built n8n workflows that automate the application steps, with a multi provider **LLM API** layer that checks model output against a schema.

Atlas, AI Powered Commercial Intelligence Workbench | GitHub
Python* | *SQL* | *PostgreSQL* | *LangGraph* | *FastAPI* | *Next.js* | *TypeScript* | *Redis* | *Docker* | *GitHub Actions

- Developed a **LangGraph analytics agent** with conversational context, SQL validation, restricted database access, query timeouts and evidence backed responses, over a PostgreSQL star schema of 24,576 synthetic pharmaceutical sales records.
- Built forecasting and analytics that reached 1.25% forecast MAPE versus 4.70% for a seasonal naive baseline, with eight data quality checks in a reproducible ETL.
- Delivered a Next.js dashboard with interactive charts and SQL inspection, and verified the Docker stack and **19 backend tests** with GitHub Actions checks.

ClinicVoice AI, Voice Receptionist for Clinics | GitHub Jun 2025 to Sep 2025
Python* | *FastAPI* | *Node.js* | *LangGraph* | *MCP* | *Groq* | *Amazon Bedrock* | *SQLite* | *Docker Compose* | *GitHub Actions

- Built a voice AI agent system with **LangGraph and MCP tools**, with six agents where only the FAQ agent calls an LLM, that answers patient questions, books, cancels and reschedules real appointments through Google Calendar and refuses medical advice.
- Added four safety layers and a **global safety gate** after live testing showed a gap, plus Amazon Bedrock as a second LLM provider behind one interface.
- Built an eval harness with a red team set, fixed the failures it found, and ran it in **GitHub Actions CI** on every push as an automated regression check.

TECHNICAL SKILLS

RAG and Document Intelligence: RAG, embeddings, pgvector, semantic search, hybrid retrieval, source quote checking, document parsing (PDF, DOCX, TXT), NLTK, spaCy, role based access control, prompt injection testing

LLMs and Agents: LangChain, LangGraph, MCP, tool calling, structured outputs, OpenAI, Claude, Gemini, Groq, Ollama, Amazon Bedrock, eval harness

Backend and Frontend: Python, FastAPI, Node.js, PostgreSQL, SQLite, Redis, Docker Compose, GitHub Actions CI, Git, React, Next.js, TypeScript

CS Fundamentals: Data structures and algorithms (DSA), DBMS, object oriented programming (OOPs), Python, Java, C++, Go

EDUCATION

Vellore Institute of Technology, Chennai | BTech, Computer Science, Specialization in AI and Machine Learning Aug 2023 to Sep 2027

CERTIFICATIONS AND ACHIEVEMENTS

- Won **First Prize** at the AGRHINOVA Hackathon, VIT Chennai, for a crop diagnostics platform that combines AI image classification with a multilingual RAG assistant.
- AWS Certified Developer Associate. AWS Cloud Practitioner. Google Cloud Professional ML Engineer. Microsoft Azure Developer Associate, Azure Fundamentals, ML Operations Engineer Associate. IBM Advanced Generative AI. Avery Dennison Scholarship.

LEADERSHIP AND EXTRACURRICULAR

- Vice President, Swarajya Marathi Literary Association (2025 to 2026). State Level Cricketer, Maharashtra Under 16.